

Article

Development of a Multifunctional Unmanned Boat Platform for Aquaculture Automation

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Abstract: To reduce labor costs in aquaculture and enhance the level of automated management, this study designed and developed a multifunctional unmanned boat platform (UBP) by integrating technologies such as sensors, satellite positioning, and artificial intelligence. The platform contains three major modules for data collection, underwater vision, and motion control, enabling functions like cruise path planning, water quality monitoring, identification of aquaculture products, and bait feeding. To verify its reliability and practicality, verification experiments were conducted in the aquaculture area of Lianyungang, China. The results show that the UBP can efficiently distribute feed to an area of 10,000 square meters within 20 min based on feeding points, outperforming the 47 min required for manual feeding. Over a two-month period, the weight of sea cucumbers raised by unmanned boats increased by 67.7% compared to those raised manually, with a 24.33% reduction in feed usage. Additionally, the unmanned boat reduced daily aquaculture costs from 225 RMB to 120.2 RMB, a total reduction of 46.7%. In conclusion, this platform reduces labor costs by improving aquaculture efficiency, and addresses limitations of the existing aquaculture feeding machinery in adaptability and real-time responsiveness, which can provide a feasible solution for aquaculture automation.

Keywords: unmanned boat; aquaculture; automatic cruising; water quality monitoring; target recognition



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1. Introduction

Aquaculture is a crucial pillar of China's food security. According to the FAO's 2020 report, The State of World Fisheries and Aquaculture, global fish production reached approximately 1.79×10^8 tons in 2018, with aquaculture production accounting for about 8.21×10^7 tons. China remains the world's largest fish producer and exporter, where small- and medium-sized aquaculture enterprises constitute over 90% of the sector. Aquaculture has significantly contributed to rural economic development and sustainable utilization of fishery resources in China. Efficient aquaculture practices, particularly automated aquaculture, are essential for industry advancement.

Feeding management represents a critical component in aquaculture operations, constituting a major portion of production costs [1]. Current feeding methods include manual feeding, fixed feeders [2,3], and mobile feeding machines. Aquaculture robots play a

pivotal role in monitoring fish behavior [4], enabling automated feeding, and reducing human intervention. By comparison, mobile feeding machines, represented by unmanned boats [5,6], have garnered attention due to their flexibility and ease of use, and are gradually being applied in fishery feeding. By integrating advanced sensors, control systems, communication technologies, and artificial intelligence, unmanned boats are capable of performing efficient and precise operations in complex and variable aquatic environments.

In the field of aquaculture, unmanned boats have been preliminarily applied in water quality monitoring, fishery feeding, and marine life observation. In terms of water quality monitoring [7,8], unmanned boats address issues such as inflexible monitoring locations and limited monitoring range. In terms of feeding [9,10], compared to traditional feeding methods, unmanned boat feeding technology resolves problems such as long feeding times, high labor requirements, and uneven feed distribution. In terms of marine life observation, artificial intelligence technologies, like deep learning, are applied to the identification of aquaculture subjects [11–13], providing crucial technical support for the refined management of fishery farming. However, there are also some common challenges in the widespread adoption of unmanned boats, such as high costs and difficulties in multifunctional integration.

By connecting to internet-enabled underwater sensors, the water quality in aquaculture areas can be monitored in real time, and the status of the cultured organisms can be tracked, enabling precise feeding and management. Water quality significantly influences the behavior of fish [14], and maintaining the aquatic environment within a range that supports optimal metabolic efficiency can maximize growth rates and feed conversion ratios. This approach can reduce feed consumption, minimize waste, and lower production costs. Additionally, the identification and real-time monitoring of aquatic animals are prerequisites for feeding management by aquaculture robots and for assisting farmers in decision-making. Adopting different feeding methods based on the growth stages of the cultured organisms can increase feed utilization, reduce feed waste, and minimize water pollution. Currently, manual feeding still dominates in aquaculture. Some existing unmanned feeding machines can only achieve automatic feeding functions, but struggle to monitor the water quality in aquaculture areas, overlooking the impact and changes in the aquatic environment. Moreover, relatively fixed feeding patterns may fail to account for changes in the growth stages of the cultured organisms, potentially leading to feed waste, water pollution, and other issues.

Therefore, this paper develops an integrated unmanned boat platform for aquaculture using EVA foam board hulls and embedded hardware components, including STM32, NodeMCU, and NeZha development boards. Leveraging the platform's mobility, it collects real-time water quality data (temperature, pH, TDS) and underwater imagery. By integrating the Monte Carlo simulation-based automatic cruising for fishing zones, a water environment quality evaluation system, and an AI edge computing-powered underwater vision system, the platform implements core functionalities encompassing water quality monitoring, bait feeding, and growth tracking. This UBP enables flexible feeding strategy optimization based on growth conditions, addressing existing limitations in feeding machinery regarding adaptability and real-time responsiveness.

The subsequent structure of this paper is organized as follows. Section 2 details the hardware system design of the unmanned boat platform, along with three core technological components: the Monte Carlo simulation-based automatic cruising algorithm, water environment quality evaluation system, and AI edge computing-powered underwater vision system. Section 3 evaluates platform performance through field validation experiments and analyzes the experimental results. Section 4 discusses the experimental findings and outlines potential future research directions. Section 5 summarizes the research achieve-

ments and their practical implications in aquaculture applications. The overall technical route of the article is shown in Figure 1.

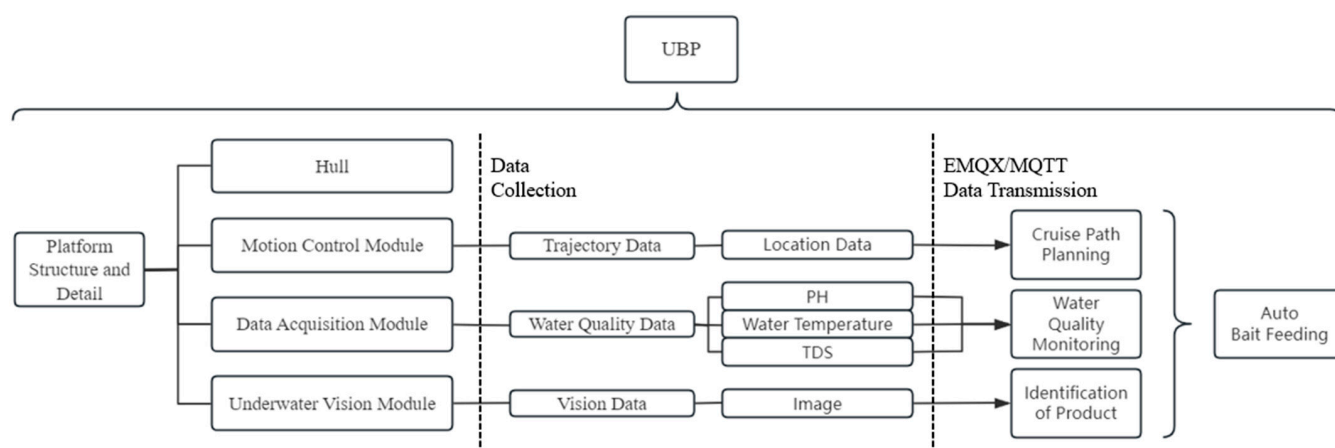


Figure 1. Overall technical route.

2. Complete Machine Structure and Working Principle

2.1. Platform Structure and Details

Aiming at the needs of small-scale aquaculture tasks, this paper develops a lightweight unmanned boat device. The hardware part mainly includes the hull, environmental data acquisition module, underwater vision system module, and hull motion control module (Table 1). Among them, the hull is made with EVA foam boards, which is not only lightweight but ensures good buoyancy performance. The environmental data acquisition module contains a NodeMCU development board, a digital water temperature sensor, a pH sensor, and a TDS turbidity sensor, which can realize accurate measurement and data transmission of various water quality parameters. The underwater vision module includes a NeZha development board and a 1080P high-definition underwater camera, which can realize real-time shooting and transmission of underwater targets. The motion control module includes a STM32 development board and a BeiDou antenna, which is dedicated to realize the precise motion of the boat. The specific parameters of the unmanned boat are shown in Table 1.

Table 1. Parameters of unmanned boat.

Module	Norm	Instructions
Hull	Width	26 cm
	Length	48.5 cm
	Battery Capacity	5200 mAh
	Weight Capacity	2.5–5 kg
	Hull Weight	1.5 kg
Environmental Data Acquisition Module	Endurance	4–5 h
	NodeMCU Development Board ¹	Operating module ESP8266; operating voltage 5 V; input voltage 5 V; digital input and output pins 10 (all can be used as PWM pins); PWM pins 10; analog input pins 1; Wi-Fi standard 802.11 b/g/n; operating mode STA/AP/STA + AP.
	DS18B20 Digital Water Temperature Sensor ²	Operating range (temperature) −55~+125 °C; Temperature accuracy ± 0.5 °C
	PH4502C PH Sensor ²	Detectable Concentration Range: PHO-14; Detectable Temperature Range: 0–80 °C; Response Time: ≤5 s; Operating Temperature: −10~50 °C (Standard Temperature 20 °C); Operating Humidity: 95% RH (Standard Humidity 65% RH)
	TDS Turbidity Sensor ²	Operating range (TDS) 0–1000 ppm; TDS accuracy ± 5% F.S. (25 °C)

Table 1. Cont.

Module	Norm	Instructions
Underwater Vision Module	NeZha Development Board ³	85 mm × 56 mm; Intel® processor N97 (Alder Lake-N); quad-core SoC clocked at 3.60 GHz, TDP of 12 W; 8 GB of LPDDR5 system memory, up to 64 GB of eMMC storage, onboard TPM 2.0, 40-pin GPIO connector, and support for Windows and Linux operating systems
	Underwater Camera ⁴	1080 P HD pixels; IP67 rated waterproof; 25 FPS image transfer rate
Motion Control Module	STM32 Development Board ²	Model: STM32F103C8T6; Core: ARM Cortex-M3; Main Frequency: 72 MHz RAM; 20 K (SRAM); ROM: 64 K (Flash); Supply Voltage: 2.0~3.6 V (Standard 3.3 V); Package: LQFP48 (48 pins)
	BeiDou Antenna ² Hull Steering ²	GPS/BDS Compatible Voltage: 7.4V; Rotation speed: 9000 r

¹ Manufactured by Espressif Systems, Shanghai, China; Purchased via Taobao; ² Manufactured by DFRobot, Shanghai, China; Purchased via Taobao; ³ Manufactured by Allwinner Technology, Zhuhai, China; Purchased via Taobao; ⁴ Manufactured by 3RD EYE, Shenzhen, China; Purchased via Taobao.

2.1.1. Hull Structure

The hull structure is the foundation of the unmanned boat. Its design directly affects the stability and reliability of the platform. In aquaculture, unmanned boats need to operate in various water conditions. These conditions include calm ponds and open waters with wind and waves. A well-designed hull must ensure stability and buoyancy in the water. It must also consider the durability for long-term use and maintenance costs. This is especially important for small-scale farmers, where cost control and operational simplicity are critical. Based on these considerations, the hull is fabricated using EVA foam boards. EVA foam boards are inexpensive, have good buoyancy, and are corrosion-resistant [15]. These properties make EVA foam boards suitable for long-term use in seawater environments. The overall hull design is streamlined to reduce water resistance. This improves sailing speed and stability. The bottom of the boat has chamfered corners. This minimizes lateral tilting and rocking in waves, enhancing the stability of the hull.

The hull structure is divided into two layers. The upper layer is the sensor layer. It mainly houses development boards and sensors required for the underwater vision module and the environmental data acquisition module. This includes the NodeMCU development board, the NeZha development board, the pH sensor, the temperature sensor, the TDS turbidity sensor, and the underwater camera. The lower part of the hull includes the bait chamber and the motion control module. The bait chamber is placed in the front for adding and dispensing bait. The motion control module is installed in the rear for powering and controlling the boat's movement. To improve airtightness and waterproofness, all electronic devices on the lower level are installed in waterproof boxes. Additional protection is provided by sealing strips. Batteries and other key components are installed at the bottom of the boat. This lowers the center of gravity and improves stability in the water. The specific placement structure is shown in Figure 2.

The functional modules of the UBP can be divided into the following seven parts: perception, planning, control, interaction, communication, payload, and navigation (Figure 3). Among them, the perception module is used to collect data from underwater sensors, such as water temperature, pH value, and TDS value. Underwater cultures are recognized through underwater cameras, combined with computer vision and machine learning. The navigation module provides real-time position estimation and trajectory planning based on GPS, BDS, and other technologies. The planning module is based on the Monte Carlo simulation to plan the cruise path and evaluate the water quality in the aquaculture area. The control module is based on the generation of low-level control commands to ensure navigation stability. The communication module is used for data exchange between the unmanned boat and the user and other devices. The interaction module carries out

human–vessel interaction through the visualization platform on the software side. The payload module is used to carry mission-specific equipment (e.g., cameras, sensors, etc.).

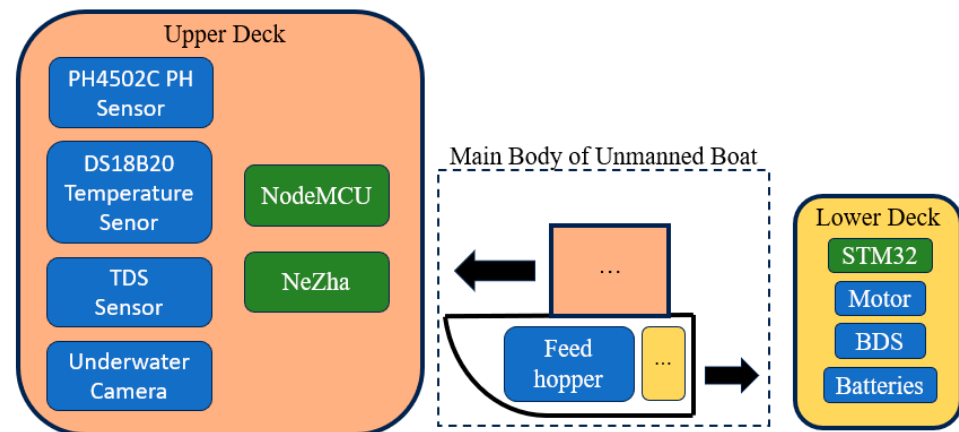


Figure 2. Hardware structure.

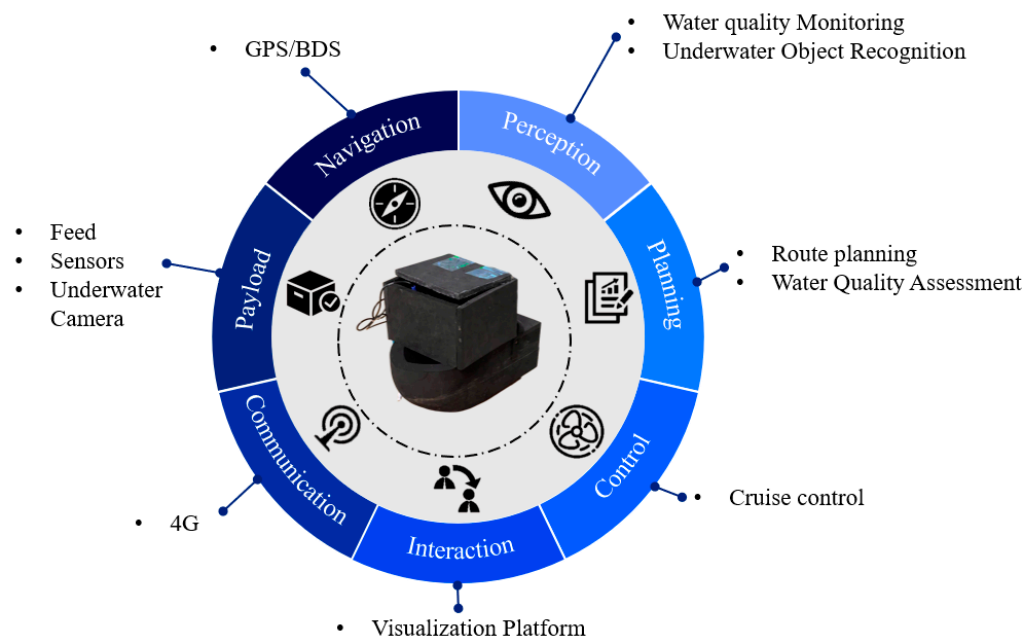


Figure 3. UBP function module.

2.1.2. Environmental Data Acquisition Module

Water quality monitoring in aquaculture is a key component to ensure the healthy growth of cultured animals. Traditional water quality monitoring usually relies on manual sampling and laboratory analysis, which is time-consuming and laborious, and difficult to obtain data in real time. The environmental data acquisition module of the unmanned boat can realize real-time and continuous water quality monitoring, providing timely and accurate information for aquaculture management. In addition, the modularized design [16–18] makes the platform easy to expand and maintain, adapting to the needs of different farms. The environmental data acquisition module is mainly developed based on the NodeMCU development board, carrying water temperature monitoring sensors, pH monitoring sensors, TDS sensors, and so on, as shown in Figure 4.

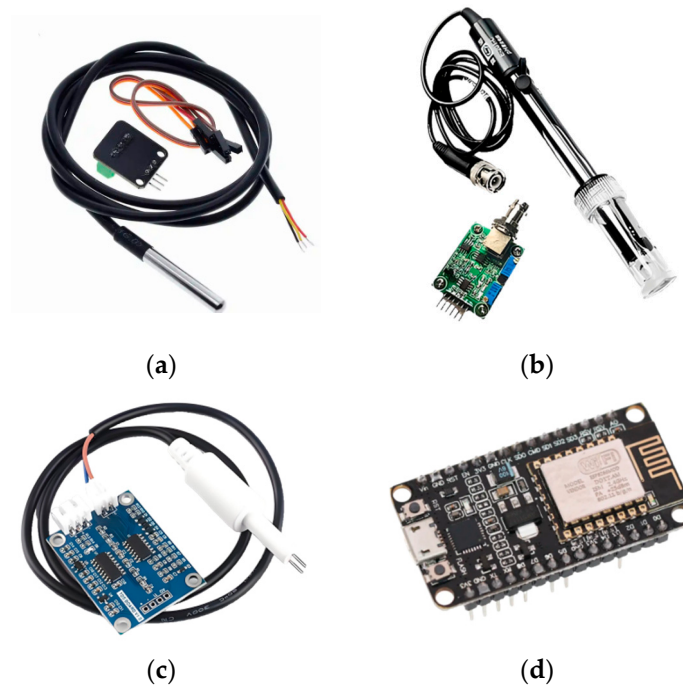


Figure 4. Sensors. (a) DS18B20 digital water temperature sensor; (b) PH4502C pH sensor; (c) TDS turbidity sensor; (d) NodeMCU development board.

(1) DS18B20 Digital Water Temperature Sensor

The DS18B20 Digital Water Temperature Sensor is a commonly used digital temperature sensor based on single bus communication technology. It features an internal integration of the temperature sensing elements and analog-to-digital conversion circuitry, capable of converting temperature changes into digital signal output. Its core is a negative temperature coefficient thermistor. As the temperature changes, the resistance value varies, and this change is converted into a digital signal through internal circuitry processing. The specific object is shown in Figure 4a.

(2) PH4502C pH Sensor

The PH4502C pH electrode sensor is mainly composed of two parts: the glass membrane electrode and the reference electrode. The glass membrane electrode contains a sensitive membrane that is selective to hydrogen ions. When the electrode is immersed in a solution, the sensitive membrane reacts with hydrogen ions in the solution, generating a potential difference proportional to the hydrogen ion concentration (i.e., pH value). This potential difference is converted into a current or voltage signal. The signal is then amplified and processed through electronic circuits, ultimately outputting an electrical signal that corresponds to the pH value of the solution. The object is shown in Figure 4b.

(3) TDS turbidity sensor

The TDS turbidity sensor is mainly used to measure the conductivity of the water solution to indirectly calculate the TDS value. The more ions dissolved in the water, the stronger its conductivity. The sensor has two electrodes, when the sensor probe is placed into the water, a stable AC voltage is applied at both ends of the electrodes, and the ions in the solution move directionally under the action of the electric field, thus generating a current. According to Ohm's law, the resistance or conductivity of the solution can be calculated by measuring the current. Since there is a certain proportionality between the TDS value and the conductivity, the TDS value can be further calculated. The physical object is shown in Figure 4c.

(4) NodeMCU development board

The NodeMCU is an open source internet of things (IoT) development board based on the ESP8266 chip [19,20], which combines Wi-Fi functionality with a programmable microcontroller to enable developers to rapidly build and deploy IoT projects. Raspberry Pi type development boards consume too much power, while ordinary microcontrollers lack native Wi-Fi capabilities. Therefore, in this paper, we choose the NodeMCU development board with lower power consumption and native Wi-Fi capability. At the heart of the development board is the ESP8266, which supports the IEEE 802.11 b/g/n standard, is capable of communicating over the 2.4 GHz band, and features low power consumption for battery-powered devices. The NodeMCU not only provides multiple GPIO pins to support digital input/output, analog inputs, PWM, I2C, SPI, and UART communications, but has an integrated Voltage converter for 3.3V and 5V peripherals. It supports multiple programming languages. The physical object is shown in Figure 4d.

2.1.3. Underwater Vision Module

The underwater vision system is an important part of the UBP to realize accurate feeding. Traditional underwater monitoring methods usually rely on manual observation and specialized equipment, which is costly and inefficient. The unmanned boat is equipped with an underwater vision module, enabling it to capture real-time photos of the underwater environment during cruising. Through AI algorithms, it identifies and monitors farmed objects, providing visualization data of the farming area to help farmers better manage the farming process. The underwater vision module is mainly developed by the NeZha development board, equipped with an underwater camera and an underwater flashlight for enhancing the underwater shooting effect. The underwater vision module is shown in Figure 5.

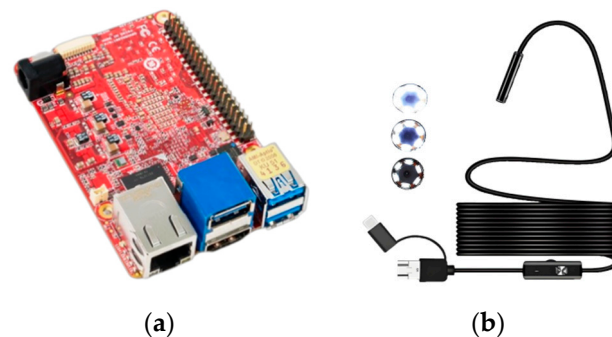


Figure 5. Underwater vision module. (a) NeZha development board; (b) underwater camera.

(1) NeZha Development Board

The NeZha development board is not only a high-performance development platform designed for internet of things (IoT) applications, but is known for its powerful AI computing power [21]. Based on a high-performance chip, the NeZha development board integrates advanced AI gas pedals that support a variety of deep learning frameworks, such as TensorFlow Lite and Caffe, enabling real-time AI inference and data processing on edge devices. Its rich hardware interfaces, such as GPIO, I2C, SPI, and UART, combined with 3.3 V and 5 V peripheral support, enable the NeZha to connect to all kinds of sensors and smart devices for all kinds of AI-driven projects. Supporting multiple programming languages, the NeZha offers a full range of capabilities from basic control to complex algorithm deployment. Its active development community and comprehensive AI repository make it ideal for IoT and AI convergence applications, greatly simplifying the development and deployment process of AI applications. The physical object is shown in Figure 5a.

(2) Underwater camera

The camera used in this paper is a 3-in-1 HD endoscope. It has multi-system compatibility and can be used on Android, Windows, and Mac systems. It has IP67 level waterproof capability, which can prevent dust from entering and short time immersion in water, and is suitable for underwater aquaculture area environment. The camera has 1080 P HD pixels with an enhanced graphics processing chip with a 25 FPS image transfer rate. There is an underwater lighting function in the head of the camera, which can enhance the underwater vision. The physical object is shown in Figure 5b.

2.1.4. Motion Control Module

The hull motion control module is the basis for unmanned boats to realize automatic cruising and accurate feeding. Traditional aquaculture equipment usually relies on fixed-position mechanical operation with low flexibility and automation. Through precise motion control, the unmanned boat can navigate autonomously in complex water environments. It follows a predetermined path for water quality testing and feeding, significantly improving work efficiency and accuracy. The boat motion control module mainly consists of an STM32 development board, a BeiDou positioning antenna, a servo, and a motor (Figure 6).

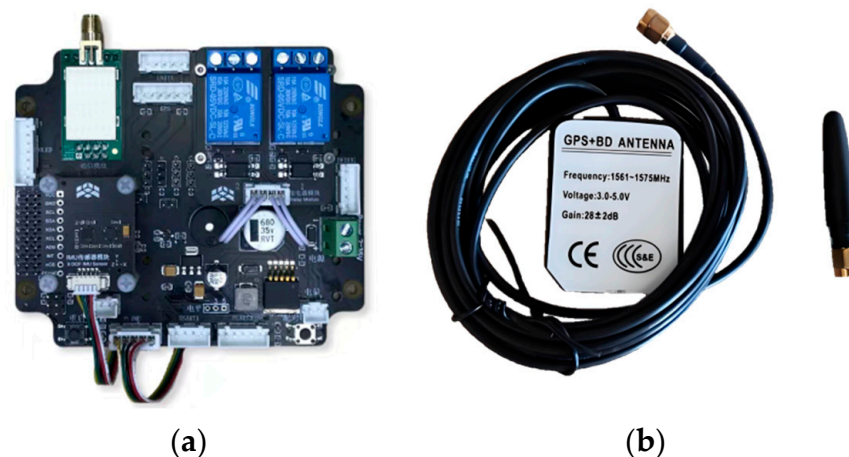


Figure 6. Motion control module. (a) STM32; (b) Beidou antenna.

(1) STM32

The STM32 development board is a microcontroller platform based on ARM Cortex-M cores, widely recognized for its versatility, cost-effectiveness, and reliability. Its low power consumption and high processing efficiency make it ideal for embedded systems, while its extensive community support and comprehensive documentation simplify development. These features make the STM32 development board a common, economical, and reliable choice for microcontroller-based projects (Figure 6a).

The hull motion control module is developed based on the STM32 development board, which controls the servo and motor through PWM signals [22,23]. The main component of the servo is the servo motor, and the so-called servo is to obey the signals and move. Before the signal comes, the rotor stops moving; after the signal comes, the rotor moves immediately. Therefore, we can input different signals to the servo to control its rotation to different angles. The servo receives a PWM signal, when the signal enters the internal circuit to generate a bias voltage, the contactor generator drives the potentiometer through the reduction gear to move, so that when the voltage difference is zero, the motor stops, so as to achieve the effect of servo. Simply put, to give the servo a specific PWM signal, the servo can rotate to the specified position.

(2) BeiDou Antenna

The BeiDou antenna is a positioning and navigation module designed for development boards, which can receive signals from the BeiDou satellite system and realize a high-precision positioning function (Figure 6b). Compared to other mainstream positioning systems, like GPS and GLONASS, the BeiDou system offers unique advantages, such as higher accuracy in the Asia–Pacific region and enhanced security due to its independent design. This makes it particularly suitable for applications requiring reliable and precise navigation. It supports a variety of communication interfaces, such as UART and SPI. This makes it easy to quickly integrate into various embedded systems, enabling accurate navigation and positioning.

2.2. Working Principle

2.2.1. Automatic Cruise Route Generation in Fishing Areas Based on the Monte Carlo Simulation

In fishery management, optimizing the visit order of feeding points is crucial to improve efficiency and reduce resource waste. Thus, the Monte Carlo simulation, which has been proved to be effective in route optimization [24,25], is applied for generating the automatic cruise route of the UBP. The basic idea of the Monte Carlo simulation is to generate a large number of random samples and conduct repetitive experiments. By statistically analyzing these samples, it estimates the solution to a given problem. Based on the feeding points in the aquaculture area, this paper designs a cruise path generation algorithm using the Monte Carlo simulation. This algorithm calculates the shortest cruise path in a shorter time to improve feeding efficiency. Figure 7 illustrates the flowchart of the algorithm.

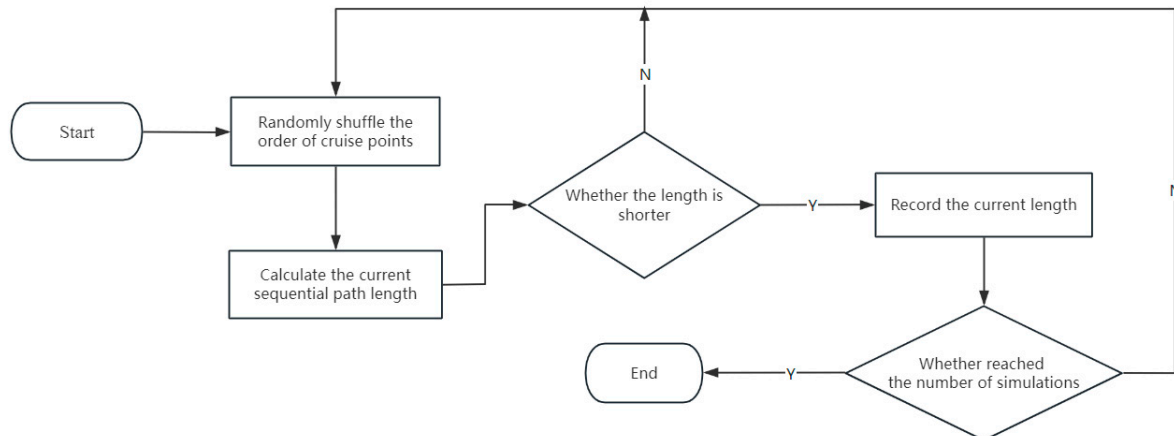


Figure 7. Flowchart of the monte carlo simulation algorithm.

2.2.2. Water Quality Evaluation System Based on Crowdsense Sensing

Since the data acquired by the unmanned boat during the cruising process is discrete point data, this paper first converts the discrete point data into continuously distributed spatial data by the method of Kriging interpolation. Secondly, this paper establishes the membership function for different types of water quality data. The membership function sets a quantitative range for each water quality indicator, transforming subjective sensory judgments into statistically and mathematically quantifiable analysis. Finally, this paper constructed a comprehensive evaluation model of water quality based on pH value, water temperature, and TDS value. The model comprehensively considered the influence of each indicator on the environmental quality of the water body. It utilized the fuzzy

comprehensive evaluation method [26–28] to assess the suitability of the fishing area, and calculated and gave the reference suggestions.

(1) Kriging interpolation

Obtaining a continuous spatial data distribution is essential in water quality monitoring and assessment. However, due to technical and resource limitations, water quality data acquisition is usually discrete. To obtain continuous water quality data within the entire fishing area, this paper employs the kriging interpolation method to acquire continuous spatial data. The kriging interpolation method has been widely applied in the fields of marine environment and water quality monitoring. Currently, there are various kriging interpolation methods, such as ordinary kriging, simple kriging, and universal kriging. Based on previous research, this paper adopts the ordinary kriging method, as shown in Equation (1), which has been validated as the most effective method in previous studies [29–31].

$$\hat{Z}(x_0) = \sum_{i=1}^n \lambda_i Z(x_i), \quad (1)$$

where $\hat{Z}(x_0)$ is the predicted value at x_0 , $Z(x_i)$ is the observed value at a known point, λ_i is the weighting factor, and n is the number of known points.

(2) Establishment of the Membership function

Water quality evaluation typically involves multiple indicators, each with different units, ranges, and levels of importance. Membership functions can unify these diverse indicators into fuzzy sets, thereby eliminating unit differences in comprehensive evaluations and facilitating unified calculation and analysis. Therefore, the membership function, as a fuzzy mathematical tool [32,33], provides an effective way to quantify the uncertainty and continuity of water quality indicators.

Based on the water quality criteria for fisheries aquaculture, this paper constructs the membership functions for three major indicators (Figure 8).

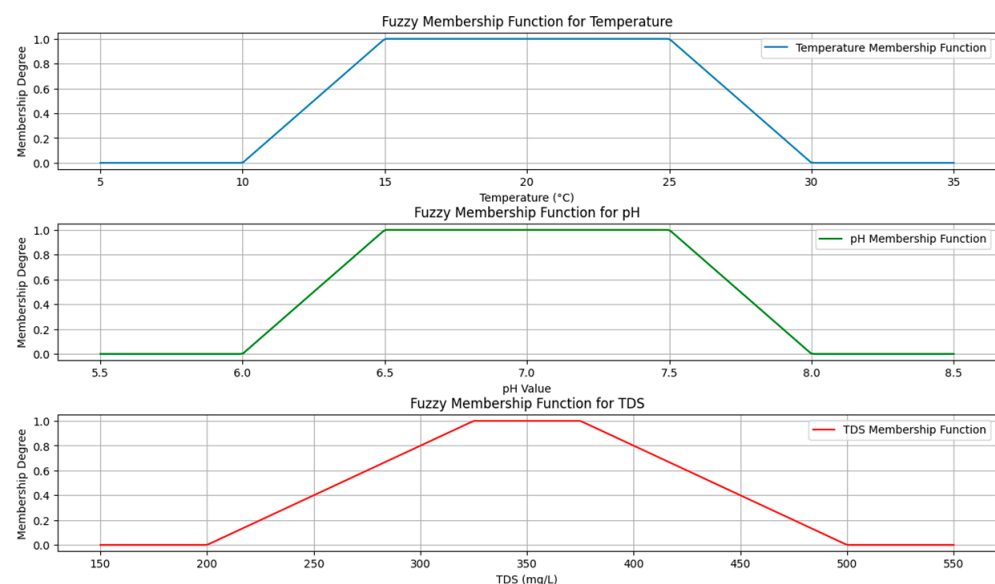


Figure 8. Indicator membership function.

(3) Comprehensive Evaluation Model

Fuzzy comprehensive evaluation is a comprehensive evaluation method based on fuzzy set theory, which mainly solves the problem of uncertainty and ambiguity in the evaluation process. It is suitable for dealing with evaluation problems that have incomplete

information or are not easy to quantify. The basic idea of fuzzy comprehensive evaluation is to use the membership function of fuzzy sets to describe the degree to which things belong to a certain category. It then conducts a comprehensive evaluation of multiple factors through fuzzy transformation, with the specific steps as follows:

Determine the set of evaluation factors: first you need to determine the factors of the evaluation object $U = \{u_1, u_2, \dots, u_n\}$, which is a collection of factors that affect the evaluation object.

Determine the set of rubrics: the set of rubrics $V = \{v_1, v_2, \dots, v_n\}$ is the set of possible evaluation results for the subject of the evaluation.

Determine the weight set: Assign weights to each factor in the factor set to form the weight set $A = \{a_1, a_2, \dots, a_n\}$, where a_i denotes the weight of factor u_i and $\sum_{i=1}^n a_i = 1$.

Perform single-factor fuzzy evaluation: Perform a fuzzy comprehensive evaluation of each factor u_i to obtain a fuzzy evaluation matrix R , where r_{ij} denotes the affiliation of factor u_i to the rubric v_j .

Fuzzy comprehensive evaluation: Through fuzzy transformation, multiply the weight set A and the fuzzy evaluation matrix R to obtain the comprehensive evaluation result $B = A \cdot R$. Here B is also a fuzzy vector, which represents the affiliation degree of the evaluation object to the set of rubrics.

Defuzzification: Finally, defuzzify the fuzzy vector B to obtain a specific evaluation result. Commonly used defuzzification methods are maximum affiliation method and center of gravity method. The formula is as follows:

$$B = A \cdot R = \{b_1, b_2, \dots, b_m\}, \quad (2)$$

where $b_j = \max_{1 \leq i \leq m} (\min(a_i, r_{ij}))$, $i = 1, 2, \dots, n$, denotes the calculation method for the j th element in the comprehensive evaluation result vector B .

2.2.3. Underwater Vision System Based on AI Edge Computing

To dynamically monitor the growing condition of aquaculture species, this paper designs an automatic detection method based on YOLOv5. Compared to previous versions of YOLO, YOLOv5 offers faster computational speed and is considered the most suitable model for real-time object detection [34].

U-Net is selected as the clarification network in this project to be used as the preprocessing of underwater aquatic images [35–37]. After processing with U-Net, the image clarity is significantly improved, the details are effectively enhanced, and the background noise is substantially reduced, providing high-quality image data support for subsequent water quality analysis and evaluation.

After image preprocessing, target detection is performed using YOLOv5. First, mosaic data augmentation, adaptive anchor box calculation, and image scaling enrich the dataset, enhancing robustness and inference speed. The backbone extracts features through the focus structure and convolutional operations, generating a $304 \times 304 \times 32$ feature map, and optimizes computational cost and accuracy using the CSP module. The neck introduces the SPP module to enhance feature fusion and expand the range of contextual features. Finally, the prediction module divides the image into different grids, predicting target bounding boxes, positions, classification results, and confidence levels to complete aquatic target detection. The YOLOv5 network structure is shown in Figure 9.

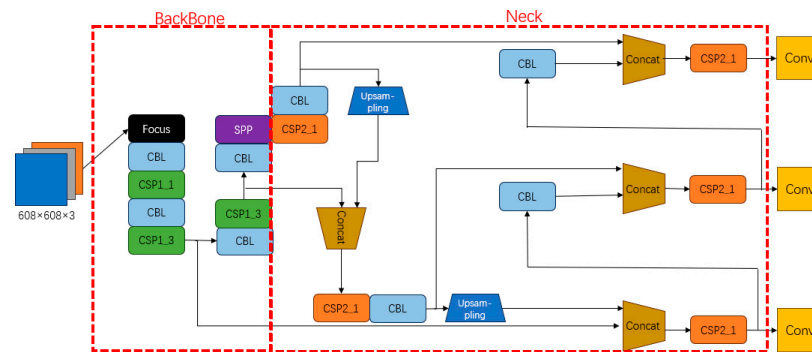


Figure 9. YOLOv5 network architecture.

3. Experimental Verification

3.1. Experimental Area and Visualization Platform

In this paper, experimental validation was conducted in Lianyungang City, Jiangsu Province. A fishery farming area measuring 100 m in length and 100 m in width was selected for testing unmanned boat-related functions. The usual environmental conditions at the test site are shown in Table 2.

Table 2. Parameters of environment.

Parameter	Measured Value	Measurement Method
Average wind speed	1–3 m/s	Handheld anemometer ¹
Ambient temperature	18–25 °C	Digital thermometer ¹
Wave height	0.1–0.3 m	Wave buoy monitoring ²
Current velocity	0.2–0.5 m/s	Doppler current meter ²

¹ Manufactured by DFRobot, Shanghai, China; Purchased via Taobao; ² Manufactured by LinkOcean, Qingdao, China; Purchased via Taobao.

Data is transmitted using the MQTT protocol [38,39], with the EMQX [40] server employed to ensure high concurrent connections and large throughput, guaranteeing the timeliness and low latency of the data. To better visualize and manage the data, we developed a visualization platform based on the Vue framework and the Leaflet framework. This platform can display the real-time operational status of the UBP and supports user functions, such as feeding point configuration. Figure 10 illustrates the data integration display interface.

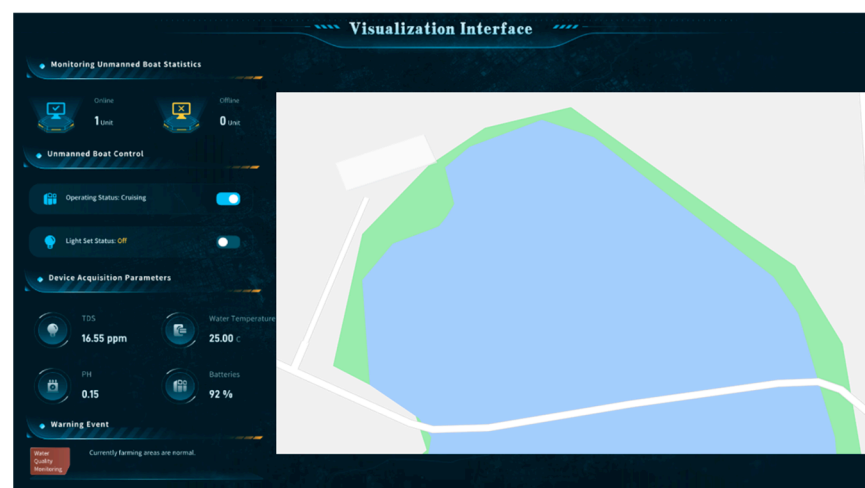


Figure 10. Visualization interface.

On the hardware side, data is transmitted to the EMQX server via the internet, where it is processed and displayed by the software side. Location data is sent in the JSON format, including a timestamp and latitude–longitude coordinates. Water quality data, which includes water temperature, pH value, and TDS value, is also sent in the JSON format, containing a timestamp and the water quality data values. The results of the AI edge detection are transmitted in the JSON format as well, including a timestamp, species, confidence level, and detection box coordinates. On the software side, by parsing the JSON data and utilizing the timestamps for matching, different types of heterogeneous data are integrated and displayed.

3.2. Cruise Function Verification

Based on the idea of the Monte Carlo simulation, for the feeding points given by the user, the shortest route is generated to complete the cruising and feeding to the fishing area. To briefly illustrate the principle, the following mathematical model is developed:

Since the fishing area is a rectangular area:

$$R = W \times H, \quad (3)$$

where W and H are the width and height of the rectangle, respectively.

There are N feeding points distributed in the following area:

$$P_i = (x_i, y_i), \quad (4)$$

where $i = 1, 2, \dots, N$.

The goal is to find a shortest path from the origin through all feeding points and back to the origin.

Define the path p as the order of the feeding points $P = \{P_{i_1}, P_{i_2}, \dots, P_{i_N}\}$, where $i_1, i_2, \dots, i_N$ is an arrangement of $1, 2, \dots, N$ and the total length of the cruise path $L(P)$ can be expressed as follows:

$$L(P) = \sum_{j=1}^{N-1} d(P_{i_j}, P_{i_{j+1}}) + d((0, 0), P_{i_1}) + d(P_{i_N}, (0, 0)), \quad (5)$$

where $d(P, Q)$ denotes the Euclidean distance between point P and point Q .

Next, Monte Carlo simulations were performed as follows:

- (1) Use a random number generator to disrupt the feeding point order $P = \{P_{i_1}, P_{i_2}, \dots, P_{i_N}\}$ to obtain a randomized feeding point order arrangement P' .
- (2) Calculate the total cruise path length $L(P')$ based on the disorganized feeding point order P' .
- (3) Compare the obtained $L(P')$ and record the shortest cruise path.

Repeat (1)–(3) until the number of simulations is reached.

As shown in Figure 11, the figure illustrates the optimal cruise paths and their corresponding path lengths obtained after 300 iterations using the Monte Carlo algorithm, the simulated annealing algorithm, and The genetic algorithm, following the selection of 10 feeding points by farmers. From this comparison chart, it is clear that the cruise path length generated by the Monte Carlo algorithm is the shortest, with the most evenly distributed spatial distribution, and a clearer path. The other algorithms generate optimal paths with issues such as crossing and overlapping within a short time frame, indicating slower optimization speeds.

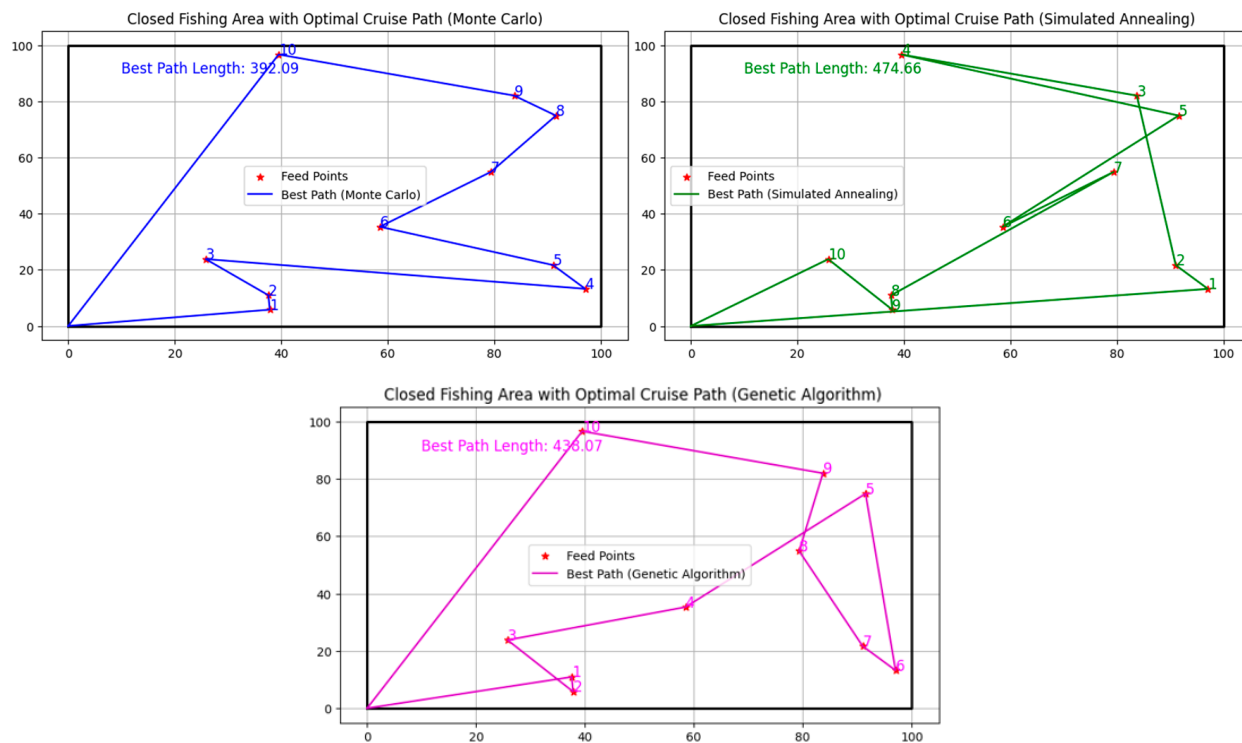


Figure 11. Optimal path after generation.

Subsequently, we conducted a quantitative analysis based on the number of iterations and path lengths. Figure 12 presents the iterations–best path length graph for the three algorithms. Compared to the simulated annealing algorithm and The genetic algorithm, the Monte Carlo method demonstrates an extremely high convergence speed in the initial phase. It only requires approximately 50 iterations to quickly find a path length close to the optimal solution and stabilizes around 400. The rapid convergence of the Monte Carlo method not only conserves computational resources but also enhances overall optimization efficiency.

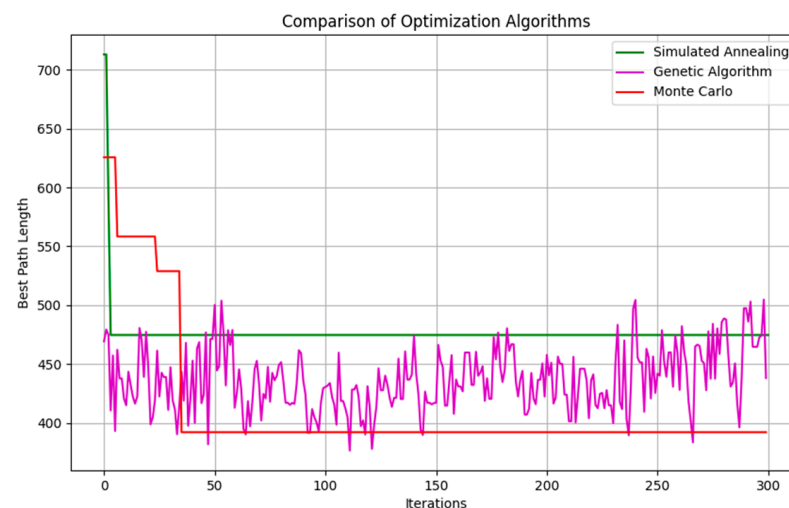


Figure 12. Iteration number–path length graph.

In contrast to the genetic algorithm, which continuously explores throughout the iteration process and may find better solutions, its significant fluctuations and longer convergence time can introduce unnecessary complexity and delays in practical applications. The cruise path generation algorithm based on the Monte Carlo simulation can rapidly find

the optimal path within a short period. This showcases the characteristics of fast generation speed and high generation efficiency.

3.3. Evaluation of Water Quality in Aquaculture Areas

According to different cruising paths, the unmanned boat carries out water quality data during cruising to obtain multiple sampling points. Seventy water quality sampling points were established in the aquaculture area to collect pH values, water temperature, and TDS values. Using the ordinary kriging method to interpolate, one can obtain the distribution of water quality in the aquaculture area. Based on the pH value, temperature, and TDS affiliation functions established in the previous section, a fuzzy comprehensive evaluation can be conducted using the kriging interpolation results. Figure 13 shows the distribution of the kriging interpolation results and the values of the membership function for pH, water temperature, and TDS values.

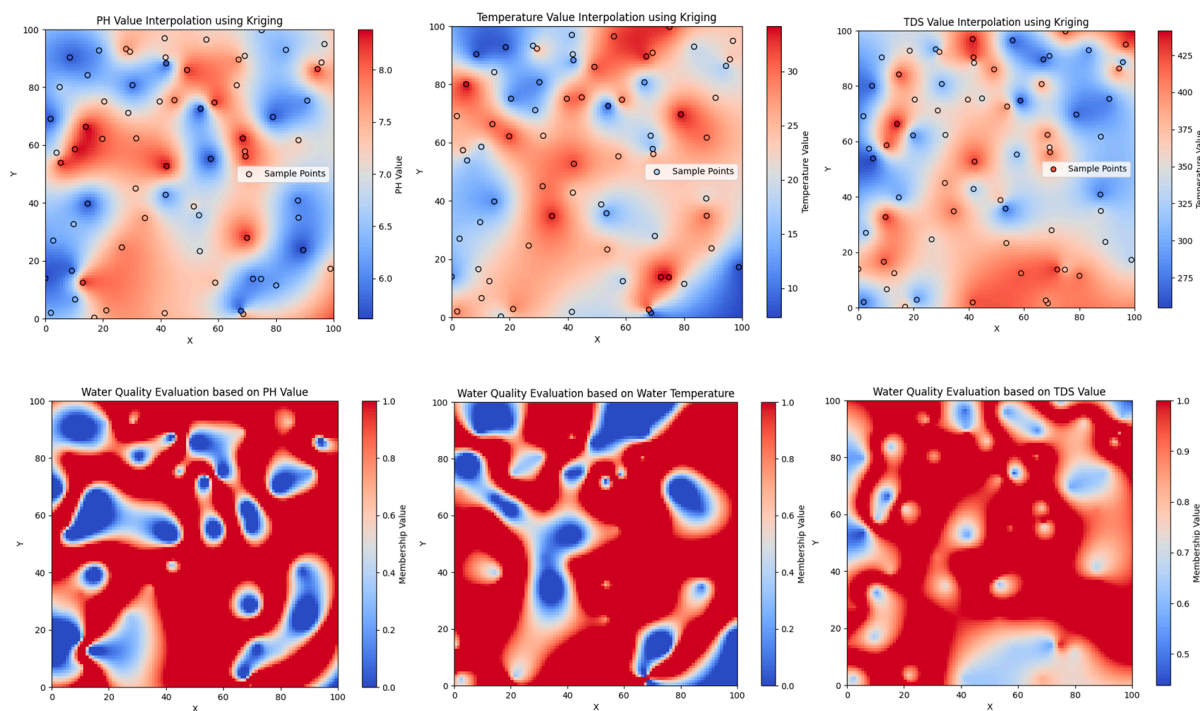


Figure 13. Ordinary kriging interpolation and distribution of membership functions.

The Root Mean Squared Error (RMSE) and the Root Mean Squared Percentage Error (RMSPE) of the predicted results for pH, temperature, and TDS values obtained through the kriging interpolation, compared to the actual sampling point data, are shown in Table 3.

Table 3. Water quality index with their corresponding RMSE and RMSPE.

Water Quality Index	pH	Temperature	TDS
RMSE	0.2	0.4 °C	20 ppm
RMSPE	2.67%	1.6%	2%

Based on the results of the pH, water temperature, and TDS membership functions, the distribution of water quality information evaluation within the entire fishing area can be clearly obtained (Figure 14).

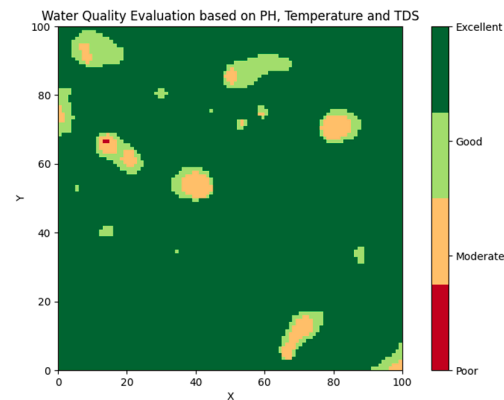


Figure 14. Results of the evaluation of fishing areas.

3.4. Recognition Results of Underwater Aquaculture

Given that the main cultured species in the test farming area are sea crabs, sea cucumbers, and scallops, among other species, this paper focuses on the training and testing of the above three cultured species based on the YOLOv5 algorithm.

In this paper, the host running system for learning is the Windows 10 operating system, intel Core i7-6800K CPU processor with a main frequency of 3.4 GHz, GTX2080Ti GPU processor, PyCharm test platform (2023.3.3 Community Edition), PyTorch machine learning framework (PyTorch 2.0.1). The whole model is trained through 80 epochs, batch is 32, and the initial learning rate is 0.0001. The dataset is derived from a curated selection of online images and the Open-Sea Farm 4K high-definition dataset. The Open-Sea Farm dataset consists of three categories (sea cucumbers, sea urchins, and scallops) and includes 2227 images [41].

The model based on the trained model works better in real operations and can accurately recognize all kinds of breeding species (Figure 15). To more accurately measure and evaluate the precision, this study selects multiple metrics for assessing object detection performance for analysis.

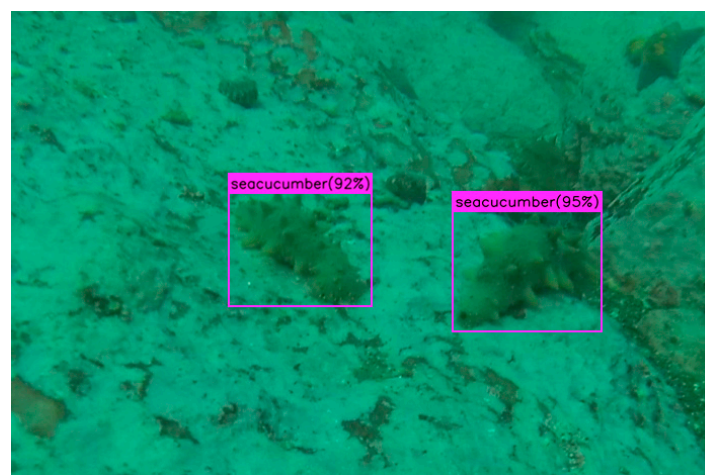


Figure 15. Underwater object recognition result.

Average Precision (AP) is a metric used to evaluate the performance of individual categories in object detection tasks. This metric combines Precision and Recall and is obtained by calculating the area under the Precision–Recall curve.

Mean Average Precision (mAP) is a core metric used to evaluate the overall performance of a model in object detection tasks. It calculates the mean of the Average Precision

(AP) across all categories, reflecting the model's comprehensive performance in multi-class detection. The higher the mAP, the better the model's detection performance.

Intersection over Union (IoU) is a metric used in object detection tasks to measure the degree of match between a predicted bounding box and a ground truth bounding box. It calculates the ratio of the area of overlap to the area of union between the predicted box and the ground truth box. Typically, an IoU threshold (e.g., 0.5) is used to determine whether a detection is successful, making it a key indicator for evaluating the accuracy of object detection. The closer the value is to 1, the higher the overlap between the predicted box and the ground truth box.

The mean Intersection over Union (mIoU) is the average of the Intersection over Union (IoU) across all categories. The higher the mIoU, the better the segmentation performance of the model.

The values of each metric are shown in Table 4.

Table 4. Performance metrics for aquaculture species detection.

Species	Scallop	Sea Cucumbers	Sea Crabs
AP	89.6%	87.8%	94.3%
mAP		90.5%	
IoU	0.78	0.72	0.81
mIoU		0.77	

3.5. Comparative Analysis for the Feeding Performance

3.5.1. Feeding Time Analysis

Table 5 provides the time required to complete feeding at all feeding points in a pond area of 10,000 square meters. The table showcases the mean distribution time and the standard deviation (S.D.). Feeding is mainly carried out in two ways: manual feeding and automatic feeding by unmanned boats. All the boats operate at a consistent speed of 0.5 m/s. In the manual feeding method, the feeding time ranges from 45 to 51 min. By contrast, the unmanned boat feeding method has a feeding time range of 19 to 20 min. For the manual feeding method, the average feeding time is 47 min, with a standard deviation of 4.3 min. On the other hand, the unmanned boat feeding method has an average duration of 20 min, with a significantly smaller standard deviation of just 0.2 min. This highlights the efficiency and consistency of the boat feeding method. This indicates that using unmanned boats for feeding can greatly save time and ensure more stable feeding times. The advantages of the unmanned boat feeding method in terms of average distribution time and lower variability underscore its potential in feeding operations, thereby enhancing resource utilization and productivity.

Table 5. Comparison of feeding time.

Treatment	Ranges	Manual Fed Mean	S.D.	Ranges	UBP Fed Mean	S.D.
Time (min)	45–51	47	4.3	19–20	20	0.2

3.5.2. Comparison of Feeding Effectiveness and Efficiency

Table 6 explores the effects of two feeding methods—manual feeding and unmanned boat feeding—on the growth of sea cucumbers and feed conversion rates. This paper compares the growth of sea cucumbers over two months under manual feeding and unmanned boat feeding conditions within the same aquaculture area. The results indicate that the initial weight of sea cucumbers in both manual and unmanned boat farming areas was approximately 50.5 g, showing that the initial sizes of the sea cucumbers were very similar. However, there was a significant difference in the total weight gain between the

two feeding methods. Sea cucumbers fed manually showed a total weight gain ranging from 62.2 to 68.7 g, with an average of 65.3 g and a standard deviation of 5.4 g. By contrast, sea cucumbers fed by unmanned boats had a wider range of total weight gain, between 105.3 and 112.5 g, with an average of 109.5 g and a standard deviation of 6.8 g. The unmanned boat feeding method appears to promote greater weight gain in sea cucumbers, with an increase of 67.7% compared to manual feeding, suggesting a potential growth advantage of this feeding method. The unmanned boat farming area consumed a total of 67.5 kg of feed, while the manual feeding area consumed 89.2 kg of feed. The unmanned boat farming area saved approximately 24.33% of feed compared to the manual feeding area. The unmanned boat farming method exhibits higher feed utilization efficiency.

Table 6. Comparison of sea cucumber weight.

Treatment	Initial Weight, Gram			Total Weight, Gram		
	Ranges	Mean	S.D.	Ranges	Mean	S.D.
Manual	49.7–51.2	50.5	0.7	62.2–68.7	65.3	5.4
UBP	49.2–51.3	50.4	0.6	105.3–112.5	109.5	6.8

These data indicate the potential advantages of unmanned boats over manual feeding methods in terms of sea cucumber growth and feed efficiency. Unmanned boat feeding can lead to better growth outcomes and more efficient feed utilization. These findings may hold practical significance for aquaculture, as improvements in growth conditions and feed efficiency can help increase yield and reduce production costs.

3.5.3. Cost Analysis

Table 7 lists the materials used in this project, along with the total costs. Table 8 provides a comparison between daily costs and manual methods. In 2024, the total manufacturing cost of this UBP was approximately 1431 RMB (equivalent to 200 USD), as shown in Table 6. All necessary materials were readily available in local markets. The machine is expected to have a service life of 2 years, and operating the unmanned boat requires only one worker at a daily cost of 100 RMB. By contrast, the manual feeding method incurs a daily cost of 200 RMB, requiring the employment of two workers to distribute the necessary feed across a 10,000 square meter area. By utilizing the unmanned boat in the farming area, the cost can be reduced from 225 RMB per day to 120 RMB per day, a reduction of 46.7%. This demonstrates the economic feasibility of unmanned boats in aquaculture, as they contribute to increased efficiency and higher profits over time.

Table 7. Cost of materials.

Material	Quantity	Unit, RMB	Total, RMB
EVA Foam ¹	3	20	60
Battery ²	1	90	90
Paddles ²	2	10	20
Motor ²	1	40	40
NodeMCU	1	12	12
Water Temp Sensor	1	2	2
pH Sensor	1	25	25
TDS Sensor	1	12	12
NeZha	1	900	900
Underwater Camera	1	100	100
STM32	1	150	150
BeiDou Antenna	1	20	20
Total boat price, RMB			1431 (US \$200)

¹ Manufactured by QianYiPaoMian, Dongguan, China; Purchased via Taobao; ² Manufactured by DFRobot, Shanghai, China; Purchased via Taobao.

Table 8. Comparison of total cost: UBP vs. manual method.

Specification	UBP	Manual
Boat Price	1431	1500
Labor Cost, RMB/day	100	200
Feed Cost, RMB/day	19	25
Maintenance Costs, RMB/day	1.2	-
Total Cost per day, RMB	120.2	225.0

4. Discussion

According to the demand for aquaculture automation and mechanization, an unmanned boat containing cruise path generation, water quality detection, and aquaculture monitoring was designed. Sailing experiments demonstrated that the unmanned boat designed in this paper based on the Monte Carlo simulation method has a faster cruise path generation speed and can find the approximate optimal path in a shorter time. Additionally, the maximum deviation of the boat's sailing does not exceed 1.5 m.

The cruising path of the boat is stable, and the positioning accuracy of the monitoring points is greatly affected by the water current and wind. This result demonstrates that the unmanned boat can cruise more accurately along the predetermined path in practical applications. However, under complex environmental conditions, such as waters with strong winds and waves, the positioning accuracy may decrease, necessitating further optimization for wind and wave resistance.

This paper assumes that the wind-resistant performance can be strengthened by improving the hull structure. By adding a V-shaped bottom to the existing chamfered design, the wave impact decomposition angle can be increased from 30° to 45°, which is expected to reduce the transverse rocking angle by 20–30%. At the same time, removable EVA fins can be added to both sides of the hull to reduce the risk of hull pitching.

Water quality monitoring system tests show that the unmanned boat can work stably during 5 to 10 s of normal navigation. The parameter accuracy of the industrial control board, the NodeMCU development board, and the sensor module meets the water quality monitoring requirements. Specifically, the measurement error of the water temperature sensor is within ± 0.5 °C, the measurement error of the pH sensor is within ± 0.1 pH, and the measurement error of the TDS sensor is within $\pm 1\%$. The developed cloud terminal real-time monitoring system works stably and is able to return real-time online data from sensors, such as pH, temperature, and dissolved oxygen of the water body, realizing real-time data visualization. This allows farmers to keep abreast of the water quality conditions in the aquaculture area and take timely measures to avoid the impact of water quality problems on aquaculture animals.

The underwater vision system is based on the fusion of SKNet and YOLOv5 aquaculture detection methods and, in actual operation, the recognition accuracies for sea crabs, sea cucumbers, and scallops are 94.3%, 87.8%, and 89.6%, respectively. Although this recognition accuracy meets the practical application requirements to a certain extent, there is still room for improvement. The recognition accuracy can be further improved by adding more training data and optimizing the model structure, especially to enhance the robustness of the model under complex background and lighting conditions.

At the same time, this study also encountered many challenges during its execution. For example, abnormal data may be generated on the hardware side of the unmanned boat (such as water quality values exceeding reasonable ranges or invalid coordinates), requiring the development of anomaly detection and handling mechanisms on the software side to prevent erroneous data from affecting system operations. When storing heteroge-

neous data from multi-source sensors, it is necessary to match and call databases storing different data.

5. Conclusions

The integrated unmanned boat platform designed in this paper for aquaculture adopts a modular design, with better compatibility between sensors and development boards, supporting multiple sensor models and achieving lower implementation costs. The hull material is highly malleable, cost-effective, and easy to disassemble and improve.

Compared to existing unmanned systems in the aquaculture industry, which often suffer from complex structures and high deployment costs, the UBP proposed in this study achieves a lightweight design, with the entire unit weighing less than 10 kg, making it more convenient for small-scale farmers to use. Additionally, it is equipped with a crowdsourced sensor array, integrating multi-source sensing modules, such as water quality monitoring and BeiDou positioning, covering daily aquaculture functions. The cost of this unmanned boat system is approximately 1431 RMB, which is only 20–40% of the current market price for unmanned boats. Compared to manual feeding, using the unmanned boat for feeding can save 24.33% of the feed. In summary, the use of the unmanned boat reduces the daily breeding cost from 225 RMB to 120.2 RMB, achieving an overall reduction of 46.7%, effectively helping farmers lower labor costs in aquaculture. This UBP provides a cost-effective automated solution for small and medium-sized farmers, assisting them in reducing labor costs, improving traditional aquaculture methods, and enhancing aquaculture efficiency. It offers a feasible solution for increasing the level of mechanization and automation in aquaculture, demonstrating significant application value.

Although this study has made some progress, there are still issues that need further resolution. The experimental environment in this study was relatively favorable, with minimal impact from factors such as wind speed on navigation accuracy. Therefore, future research will focus on enhancing the unmanned boat's resistance to wind and waves in complex environments. Algorithms for wind and wave resistance will be designed to improve navigation accuracy under challenging conditions. Additionally, hull design and structure will be further refined, testing different materials and hull shapes to enhance the stability of the unmanned boat in complex water environments. Secondly, the robustness of underwater recognition algorithms under complex backgrounds and lighting conditions needs further improvement. This will be achieved by increasing the diversity of training datasets and expanding the adaptability of algorithms to enhance their robustness under varying lighting and background conditions.

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